

# Methods 3: Multilevel Statistical Modeling and Machine Learning

Week 40: *Evaluating generalised linear mixed models — and the spectre of collinearity:*

September 29, 2026

# The course plan

## *Week 37:*

Lesson 0: What is it all about?

Class 0: Setting up R and Python — and recollection of the general linear model

## *Week 38:*

Lesson 1: Multilevel linear regression

Class 1: Modelling subject level effects – and how do they differ from group level effects?

## *Week 39:*

Lesson 2: Beyond linear regression — link functions

Class 2: What to do when the response variable is not continuous?

## ***Week 40:***

**Lesson 3: Evaluating generalised linear mixed models — and the spectre of collinearity**

**Class 3: Code review**

## *Week 41:*

Lesson 4: Explanation versus prediction

Class 4: Summarising mixed models

## *Week 42:*

no teaching (go fetch potatoes instead)

## *Week 43:*

Lesson 5: Mid-way evaluation and machine learning introduction

Class 5: Getting Python running

## *Week 44:*

Lesson 6: Linear and logistic regression revisited (machine learning)

Class 6: Categorizing responses based on informed guesses

## *Week 45:*

Lesson 7: Model evaluation and hyperparameter tuning

Class 7: Code review

## *Week 46:*

Lesson 8: Dimensionality reduction — principled component analysis (PCA)

Class 8: What to do with very rich data?

## *Week 47:*

Lesson 9: Neural networks

Class 9: Code review

## *Week 48:*

Lesson 10: Unsupervised classification

Class 10: Create video explainers of central concepts (shared resource)

## *Week 49:*

Lesson 0 again: What was it all about? and final evaluation

Class 11: Summarising machine learning

## *Week 50:*

Student presentations:

Class 12: Work on portfolios: Ask anything!

# Recap – generalised modelling

- At least four ingredients needed

1) A data vector:  $y = (y_1, \dots, y_n)$

2) Predictors:  $X$  and coefficients  $\beta$ , forming a linear predictor  $X\beta$

3) A *link function*  $g$ : yielding a vector of transformed data  $\hat{y} = g^{-1}(X\beta)$   
that are used to model the data

4) A data distribution:  $p(y | \hat{y})$

$$(X\beta = \beta_0 + X_1\beta_1 + \dots + X_k\beta_k)$$

Gelman A, Hill J (2006) Data Analysis Using Regression and Multilevel/Hierarchical Models. Cambridge University Press: Chapter 6

# Learning goals and outline

## *Evaluating Generalised linear mixed models*

- 1) Learning tools for comparing models
  - 1) Variance explained
  - 2) Likelihood ratio tests
  - 3) Information criteria
- 2) Bridging to out-of-sample
  - 1) Introducing regularisation

# Why are we modelling?

To be able to  
understand the world



**EXPLANATION**

To be able to predict  
and control the world



By Silly rabbit - self-made (by User:Silly rabbit).  
Updated in the Gimp by User:Michaelrayw2.,  
CC BY 3.0, <https://commons.wikimedia.org/w/index.php?curid=115478248>

**PREDICTION/CONTROL**

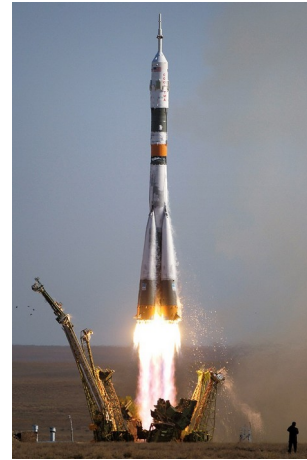
# Why are we modelling?

To be able to  
understand the world

$$F = G \frac{m_1 m_2}{r^2}$$

**EXPLANATION**

To be able to predict  
and control the world



**PREDICTION /CONTROL**

# What constitutes a good model?

Accurate estimation  
of the underlying  
parameters of the  
population distribution

Generalisation to new  
data

**EXPLANATION**

**PREDICTION/CONTROL**

Within an **explanatory** framework, how can we assess whether we have done a good job?

# 1) Variance explained

# R summary

```
##  
## Call:  
## lm(formula = mpg ~ wt, data = mtcars)  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -4.5432 -2.3647 -0.1252  1.4096  6.8727   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept)  37.2851     1.8776   19.858 < 2e-16 ***  
## wt          -5.3445     0.5591   -9.559 1.29e-10 ***  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Residual standard error: 3.046 on 30 degrees of freedom  
## Multiple R-squared:  0.7528, Adjusted R-squared:  0.7446   
## F-statistic: 91.38 on 1 and 30 DF,  p-value: 1.294e-10
```

# $R^2$ (coefficient of determination)

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \mu_y)^2}$$

# $R^2$ ; adjusted

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \mu_y)^2}$$

$$\bar{R}^2 = 1 - (1 - R^2) \frac{n-1}{n-p-1}$$

$n$ : number of observations

$p$ : number of predictors beyond the constant term

```
lm(formula = mpg ~ wt, data = mtcars)
```

Coefficients:

| (Intercept) | wt     |
|-------------|--------|
| 37.285      | -5.344 |

---

"R-squared: 0.753"

"R-squared, adjusted: 0.745"

```
lm(formula = mpg ~ I(wt^2) + wt, data = mtcars)
```

Coefficients:

| (Intercept) | I(wt^2) | wt      |
|-------------|---------|---------|
| 49.931      | 1.171   | -13.380 |

---

"R-squared: 0.819"

"R-squared, adjusted: 0.807"

---

```
lm(formula = mpg ~ I(wt^3) + I(wt^2) + wt, data = mtcars)
```

Coefficients:

| (Intercept) | I(wt^3) | I(wt^2) | wt        |
|-------------|---------|---------|-----------|
| 48.40370    | 0.04594 | 0.68938 | -11.82598 |

"R-squared: 0.819"

"R-squared, adjusted: 0.8"

```
lm(formula = mpg ~ I(wt^4) + I(wt^3) + I(wt^2) + wt, data = mtcars)
```

Coefficients:

| (Intercept) | I(wt^4) | I(wt^3) | I(wt^2)  | wt      |
|-------------|---------|---------|----------|---------|
| 14.4558     | -0.3863 | 5.2004  | -23.7018 | 36.6195 |

"R-squared: 0.822"

"R-squared, adjusted: 0.796"

# $R^2$ – multilevel models

$$R_{GLMM}(m)^2 = \frac{\sigma_f^2}{\sigma_f^2 + \sigma_\alpha^2 + \sigma_\epsilon^2} \qquad R_{GLMM}(c)^2 = \frac{\sigma_f^2 + \sigma_\alpha^2}{\sigma_f^2 + \sigma_\alpha^2 + \sigma_\epsilon^2}$$

$\sigma_f^2$ : variance of the first-level effects

$\sigma_\alpha^2$ : variance of the second-level effects

$\sigma_\epsilon^2$ : unexplained variance

$m$ : marginal variance

$c$ : conditional variance

# $R^2$ – multilevel models

Linear mixed model fit by REML ['lmerMod']

Formula: Reaction ~ Days + (1 | Subject)

Data: sleepstudy

REML criterion at convergence: 1786.465

Random effects:

| Groups | Name | Std.Dev. |
|--------|------|----------|
|--------|------|----------|

|         |             |       |
|---------|-------------|-------|
| Subject | (Intercept) | 37.12 |
|---------|-------------|-------|

|          |  |       |
|----------|--|-------|
| Residual |  | 30.99 |
|----------|--|-------|

Number of obs: 180, groups: Subject, 18

Fixed Effects:

| (Intercept) | Days |
|-------------|------|
|-------------|------|

|        |       |
|--------|-------|
| 251.41 | 10.47 |
|--------|-------|

$\hat{\sigma}_{\alpha}$

$\hat{\sigma}_{\epsilon}$

$$\hat{\sigma}_f = \text{sd}(X\hat{\beta})$$

```
library(lme4)
```

```
mm <- lmer(Reaction ~ Days + (1 | Subject), data=sleepstudy)
```

```
var.comp      <- as.data.frame(VarCorr(mm))
```

```
X <- model.matrix(mm)
```

```
beta.hat <- fixef(mm)
```

```
sigma.f      <- sd(X %*% beta.hat)
```

```
sigma.alpha  <- var.comp$sdcor[1]
```

```
sigma.epsilon <- var.comp$sdcor[2]
```

```
R.squared.m <- sigma.f^2 / (sigma.f^2 + sigma.alpha^2 + sigma.epsilon^2)
```

```
R.squared.c <- (sigma.f^2 + sigma.alpha^2) /  
  (sigma.f^2 + sigma.alpha^2 + sigma.epsilon^2)
```

```
print(R.squared.m)
```

```
## [1] 0.2798856
```

```
print(R.squared.c)
```

```
## [1] 0.7042555
```

# With slopes...

```
mm <- lmer(Reaction ~ Days + (Days | Subject), data=sleepstudy)

unscaled.SIGMA <- getME(mm, 'Lambda') %*% getME(mm, 'Lambdat')
SIGMA <- sigma(mm)^2 * unscaled.SIGMA
Z <- getME(mm, 'Z')
n <- length(sleepstudy$Reaction)
X <- model.matrix(mm)
beta.hat <- fixef(mm)

var.comp <- as.data.frame(VarCorr(mm))

obs.var <- Z %*% SIGMA %*% t(Z)

sigma.f <- sd(X %*% beta.hat)
sigma.alpha <- sqrt(sum(diag(obs.var)) / n)
sigma.epsilon <- var.comp$sdcor[4]

R.squared.m <- sigma.f^2 / (sigma.f^2 + sigma.alpha^2 + sigma.epsilon^2)
R.squared.c <- (sigma.f^2 + sigma.alpha^2) /
  (sigma.f^2 + sigma.alpha^2 + sigma.epsilon^2)

print(R.squared.m)
```

Johnson PCD (2014) Extension of Nakagawa & Schielzeth's R2GLMM to random slopes models. *Methods in Ecology and Evolution* 5:944–946.  
<https://doi.org/10.1111/2041-210X.12225>

```

Linear mixed model fit by REML ['lmerMod']
Formula: Reaction ~ Days + (Days | Subject)
  Data: sleepstudy
REML criterion at convergence: 1743.628
Random effects:
 Groups   Name                Std.Dev. Corr
Subject  (Intercept)  24.741
          Days         5.922    0.07
Residual                25.592
Number of obs: 180, groups:  Subject, 18
Fixed Effects:
(Intercept)          Days
    251.41         10.47

```

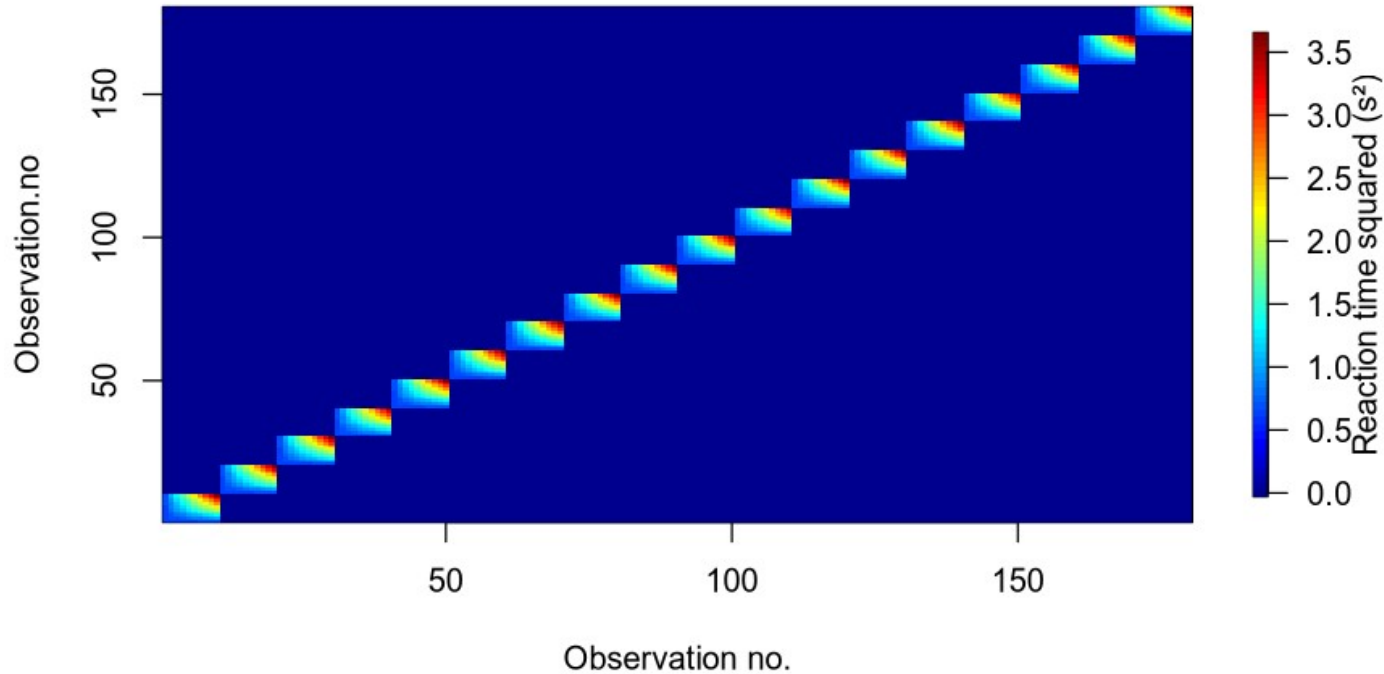
$$B \sim N(0, \Sigma)$$

$$\hat{\Sigma} = \begin{bmatrix} 24.741^2 & 9.61 \\ 9.61 & 5.922^2 \end{bmatrix}$$

$$\text{obs.var} = \mathbf{Z} \Sigma \mathbf{Z}^T$$

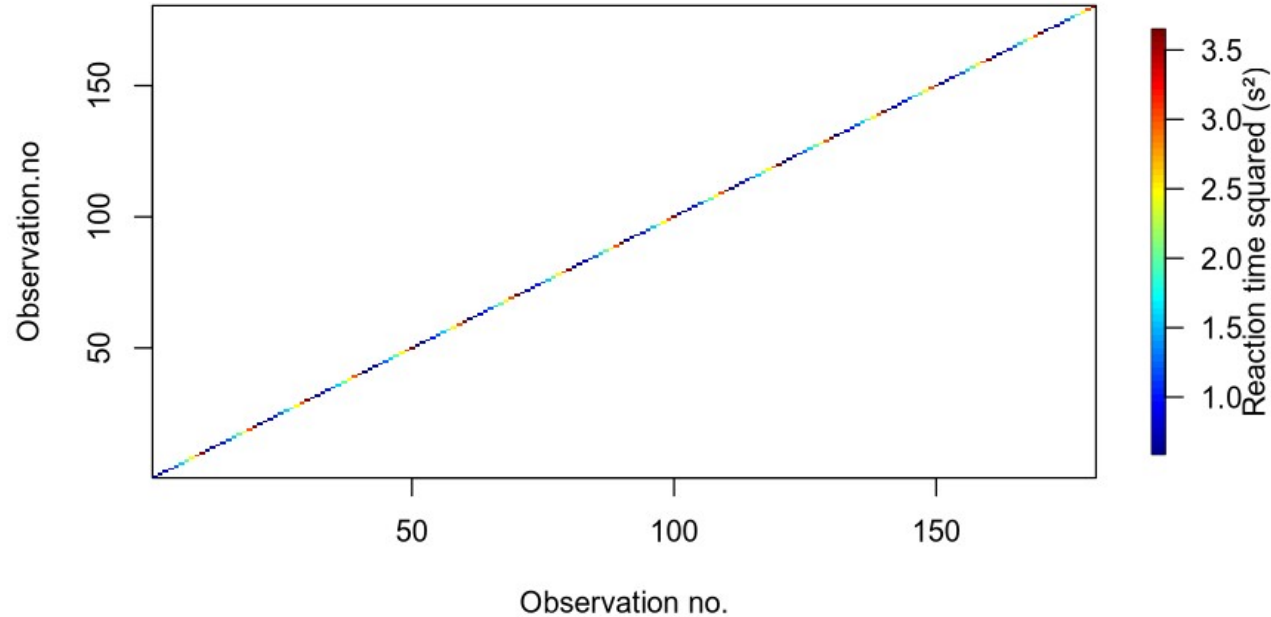
$$\sigma_{\alpha}^2 = \text{tr}(\text{obs.var})/n$$

Second-level effect variance



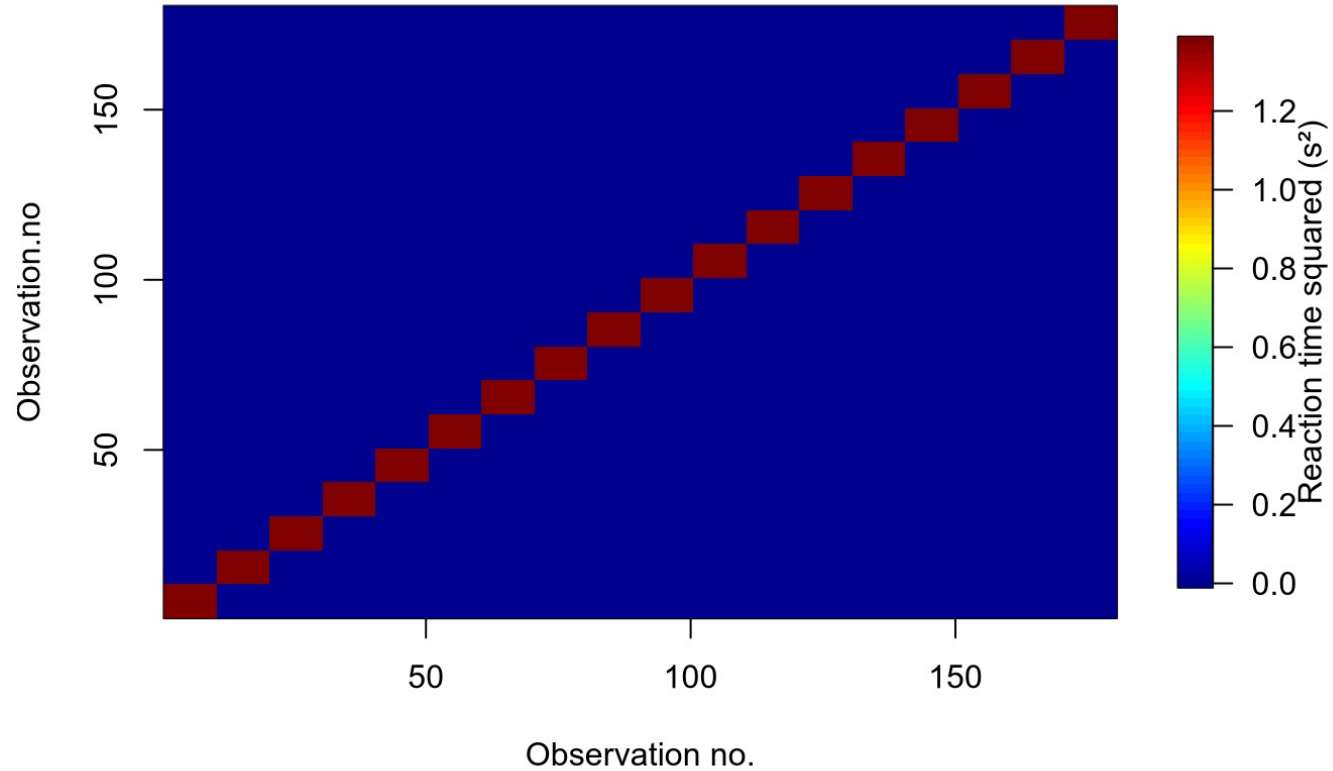
$$\sigma_{\alpha}^2 = \text{tr}(\text{obs.var})/n$$

Diagonal of Second-level effect variance



Johnson PCD (2014) Extension of Nakagawa & Schielzeth's R2GLMM to random slopes models. *Methods in Ecology and Evolution* 5:944–946.  
<https://doi.org/10.1111/2041-210X.12225>

## Second-level effect variance



Discuss: what  
Z is behind  
this matrix?

# MuMIn

## MULTI-MODEL INFERENCE

```
library(MuMIn)  
print(r.squaredGLMM(mm))
```

```
##                R2m                R2c  
## [1,] 0.2798856 0.7042555
```

# Variance explained

- Pros
  - $R^2$  is intuitive
- Cons
  - More complex models will always explain more variance
  - Hard to interpret in the case of collinearity
  - $R^2$  may not give us what we want

$$y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \varepsilon_i. \quad (1)$$



$$\hat{y}_i = 1.6 + 0.35X_{1i} + 0.62X_{2i}.$$



$$R^2 = 0.750$$



Tempting interpretation: the parameter estimates are likely to be true, because  $R^2$  is large

Correct interpretation:

“if one fits a model with the form of equation 1 in each new sample – *each time estimating new values of  $b_0$ ,  $b_1$ , and  $b_2$*  – what will be the average proportional reduction in the sum of squared errors?”  
(Yarkoni and Westfall 2017)

## 2) Likelihood ratios

# Log-likelihoods

$$l(\hat{\beta}, \hat{\sigma}^2 | y) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\hat{\sigma}^2) - \frac{n}{2}$$

$$l(\hat{\beta} | y) = \sum_{i=1}^n y_i \log(\hat{y}_i) + \log(1 - y_i)(1 - \hat{y}_i)$$

$$l(\hat{\beta} | y) = \sum_{i=1}^n y_i \log(\hat{y}_i) - \hat{y}_i - \log(y_i!)$$

```
model.ranint <- lmer(Reaction ~ Days + (1 | Subject), data=sleepstudy)
model.ranslope <- lmer(Reaction ~ Days + (Days | Subject), data=sleepstudy)
```

```
print(ll.i <- logLik(model.ranint))
```

```
## 'log Lik.' -893.2325 (df=4)
```

```
print(ll.s <- logLik(model.ranslope))
```

```
## 'log Lik.' -871.8141 (df=6)
```

Is the gain in log-likelihood worth the parameters we spend, and why is df increased by 2?

# Likelihood-ratio test

$$LR = -2(l(\theta_2) - l(\theta_1))$$

```
print(anova(model.ranint, model.ranslope))
```

refitting model(s) with ML (instead of REML)

Data: sleepstudy

Models:

model.ranint: Reaction ~ Days + (1 | Subject)

model.ranslope: Reaction ~ Days + (Days | Subject)

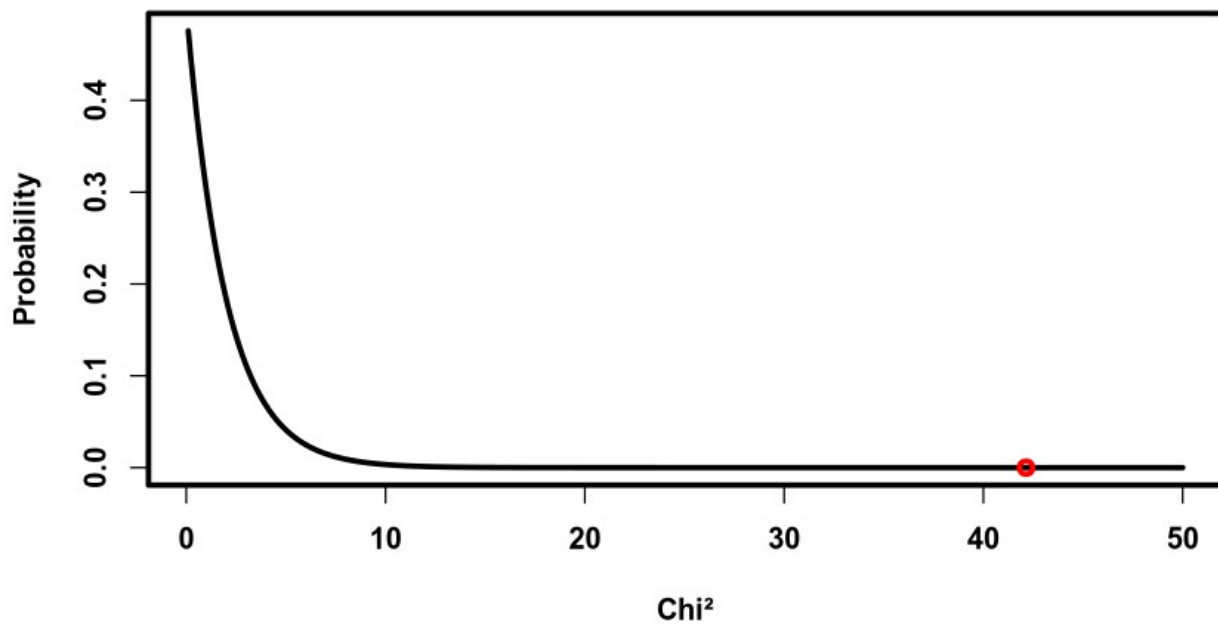
|                | npar | AIC    | BIC    | logLik  | deviance | Chisq  | Df | Pr(>Chisq)    |
|----------------|------|--------|--------|---------|----------|--------|----|---------------|
| model.ranint   | 4    | 1802.1 | 1814.8 | -897.04 | 1794.1   |        |    |               |
| model.ranslope | 6    | 1763.9 | 1783.1 | -875.97 | 1751.9   | 42.139 | 2  | 7.072e-10 *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

$$\chi^2(2)$$

PDF, df=2



$\Pr(>\text{Chisq})$

$7.072\text{e-}10$

# When is it valid?

“The LR test is only adequate for testing fixed effects when both the ratio of the total sample size to the number of fixed-effect levels being tested and the number of random-effect levels (blocks) are large. *We have found little guidance and no concrete rules of thumb in the literature [...]*” (my highlights, Bolker et al., 2009)



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# Journal of Memory and Language

journal homepage: [www.elsevier.com/locate/jml](http://www.elsevier.com/locate/jml)



## Random effects structure for confirmatory hypothesis testing: Keep it maximal



Dale J. Barr<sup>a,\*</sup>, Roger Levy<sup>b</sup>, Christoph Scheepers<sup>a</sup>, Harry J. Tily<sup>c</sup>

<sup>a</sup>*Institute of Neuroscience and Psychology, University of Glasgow, 58 Hillhead St., Glasgow G12 8QB, United Kingdom*

<sup>b</sup>*Department of Linguistics, University of California at San Diego, La Jolla, CA 92093-0108, USA*

<sup>c</sup>*Department of Brain and Cognitive Sciences, Massachusetts Institute of Technology, Cambridge, MA 02139, USA*

“Through theoretical arguments and Monte Carlo simulation, we show that LMEMs generalize best when they include the **maximal random effects structure justified by the design**. [...] Maximal LMEMs should be the ‘gold standard’ for confirmatory hypothesis testing in psycholinguistics and beyond. (my highlight)”

Barr DJ, Levy R, Scheepers C, Tily HJ (2013) Random effects structure for confirmatory hypothesis testing: Keep it maximal. *Journal of Memory and Language* 68:255–278.  
<https://doi.org/10.1016/j.jml.2012.11.001>

| $N_{items}$                                                  | Type I      |             | Power |      |
|--------------------------------------------------------------|-------------|-------------|-------|------|
|                                                              | 12          | 24          | 12    | 24   |
| <i>Type I: Error at or near <math>\alpha = .05</math></i>    |             |             |       |      |
| min- $F'$                                                    | .027        | .031        | .327  | .512 |
| LMEM, maximal, $\chi^2_{LR}$                                 | <u>.059</u> | <u>.056</u> | .460  | .610 |
| LMEM, no random correlations, $\chi^2_{LR}{}^a$              | .059        | .056        | .461  | .610 |
| LMEM, no within-unit intercepts, $\chi^2_{LR}{}^a$           | .056        | .055        | .437  | .596 |
| LMEM, maximal, $t$                                           | .072        | .063        | .496  | .629 |
| LMEM, no random correlations, $t$                            | .072        | .062        | .497  | .629 |
| LMEM, no within-unit intercepts, $t^a$                       | .070        | .064        | .477  | .620 |
| $F_1 \times F_2$                                             | .057        | .072        | .440  | .643 |
| <i>Type I: Error far exceeding <math>\alpha = .05</math></i> |             |             |       |      |
| $F_1$                                                        | .176        | .139        | .640  | .724 |
| LMEM, no random correlations, MCMC <sup>a</sup>              | .187        | .198        | .682  | .812 |
| LMEM, random intercepts only, MCMC                           | .415        | .483        | .844  | .933 |
| LMEM, random intercepts only, $\chi^2_{LR}$                  | <u>.440</u> | <u>.498</u> | .853  | .935 |
| LMEM, random intercepts only, $t$                            | .441        | .499        | .854  | .935 |

Barr DJ, Levy R, Scheepers C, Tily HJ (2013) Random effects structure for confirmatory hypothesis testing: Keep it maximal. *Journal of Memory and Language* 68:255–278.  
<https://doi.org/10.1016/j.jml.2012.11.001>

# Likelihood ratio

- Pros
  - Models can be compared in a principled way by reference to a theoretical distribution,  $\chi^2$ . (In the single level case,  $F$  can be calculated)
- Cons
  - Models have to be nested in one another
  - Maximum likelihood fitting may be biased for complex models
  - Requires large sample sizes
  - Be careful if collinearity is high

# **3) Information criteria**

# Information criteria

$$\text{deviance} = -2l(\hat{\theta})$$

$$\text{AIC} = \text{deviance} + 2k$$

$k$ : number of predictors

When we add  $k$  predictors that are pure noise, deviance is reduced by an amount corresponding to a  $\chi^2$  distribution with  $k$  degrees of freedom.

“On average, a predictor needs to reduce the deviance by 2 in order to improve the fit to new data”

```
## Data: sleepstudy
## Models:
## model.ranint: Reaction ~ days_deprived + (1 | Subject)
## model.ranslope.and.int: Reaction ~ days_deprived + (days_deprived | Subject)
##
```

|                           | Df | AIC    | BIC    | logLik  | deviance | Chisq  | Chi | Df |
|---------------------------|----|--------|--------|---------|----------|--------|-----|----|
| ## model.ranint           | 4  | 1446.5 | 1458.4 | -719.25 | 1438.5   |        |     |    |
| ## model.ranslope.and.int | 6  | 1425.2 | 1443.0 | -706.58 | 1413.2   | 25.332 |     | 2  |

```
## Pr(>Chisq)
## model.ranint
## model.ranslope.and.int 3.156e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

# How to estimate $k$ ?

# With $J$ parameters

- Complete pooling
  - all  $J$  parameters collapsed into 1 parameter, ( $k = 1$ )
- No pooling
  - each of the  $J$  parameters is estimated, ( $k = J$ )
- Partial pooling
  - the number of effective parameters may be estimated by sampling and the *Deviance Information Criterion* can be estimated (beyond this course)
  - the number of parameters, loosely speaking, depend on whether the parameters are estimated mostly by the group average or the individual's own average ( $k = ?$ )

# Information criteria

- Pros
  - Models can be compared even though one is not nested within the other (response data has to be the same though)
- Cons
  - Number of effective parameters not well defined for multilevel models
  - Maximum likelihood fitting may be biased for complex models
  - Be careful if collinearity is high

# Interim summary

- Our different ways of comparing models seem to rely on estimating the number of effective parameters, on which there is no clear consensus
- Collinearity is a problem for all methods we have covered
  - Collinearity leads to unstable models
    - What can we do about it?

# Collinearity

```
##  
## Call:  
## lm(formula = hp ~ mpg + wt + drat + qsec, data = mtcars)  
##  
## Coefficients:  
## (Intercept)          mpg           wt          drat          qsec  
##    473.779       -2.877       26.037       4.819       -20.751
```

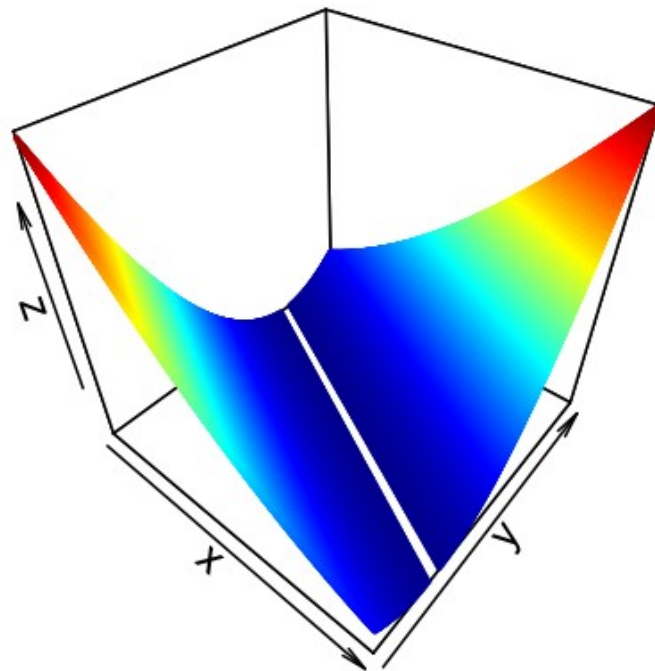
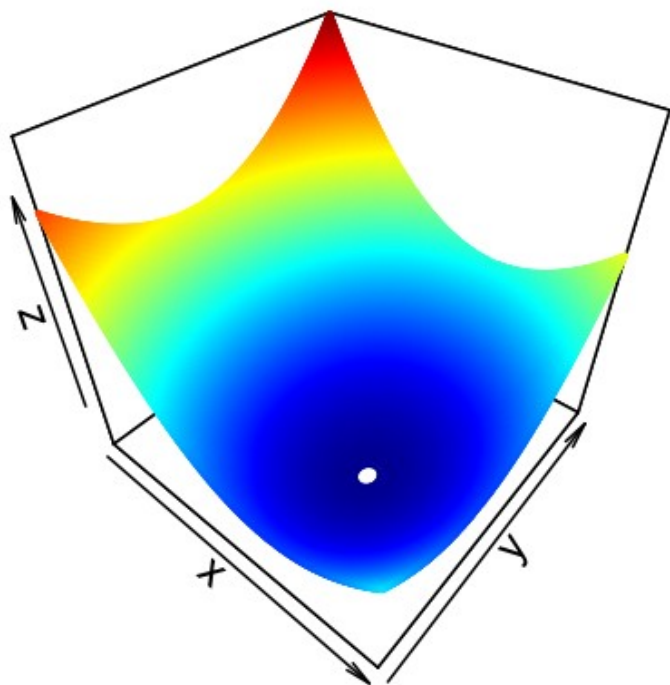
Assuming no collinearity, what is the interpretation of the coefficients?  
With collinearity, is that interpretation possible?

# Stability of a model:

Feeding new data or adding new predictor variables will not change the parameter estimates a lot

Seems to be a good ingredient for a model that generalises well

# Stability of a model



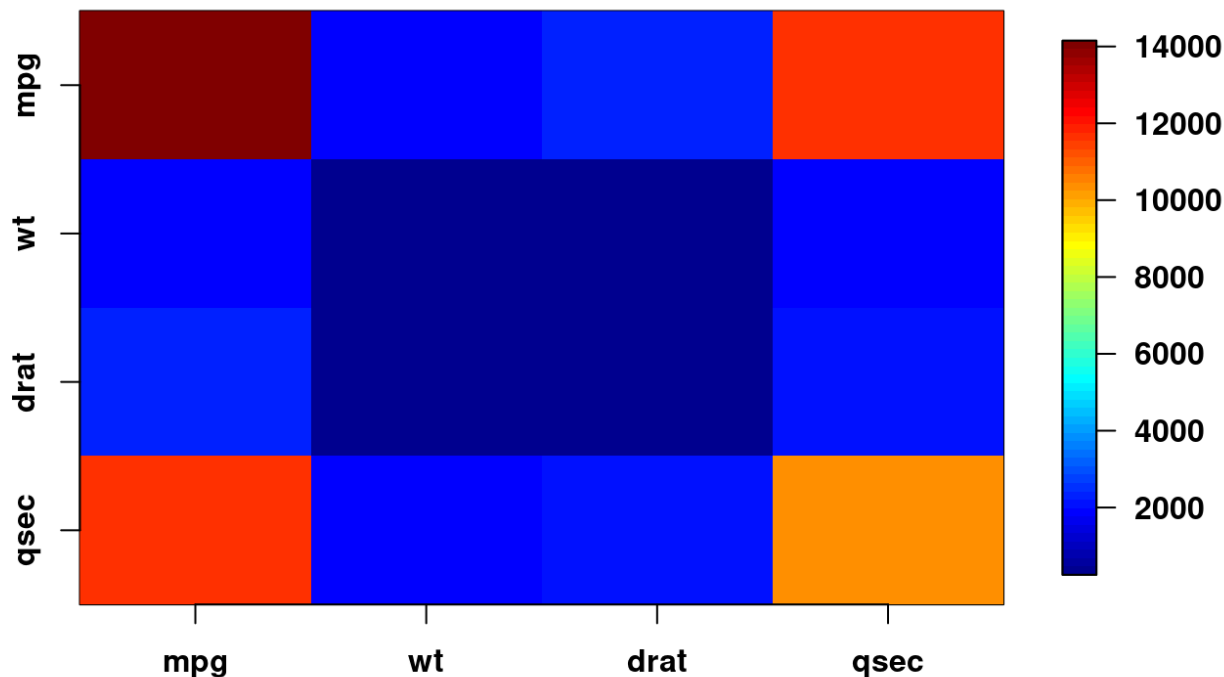
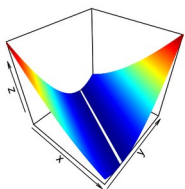
Here you see the gradient descent profiles for two models.  
Which of these models is the most stable?

CC BY Licence 4.0; Lau Møller Andersen 2026

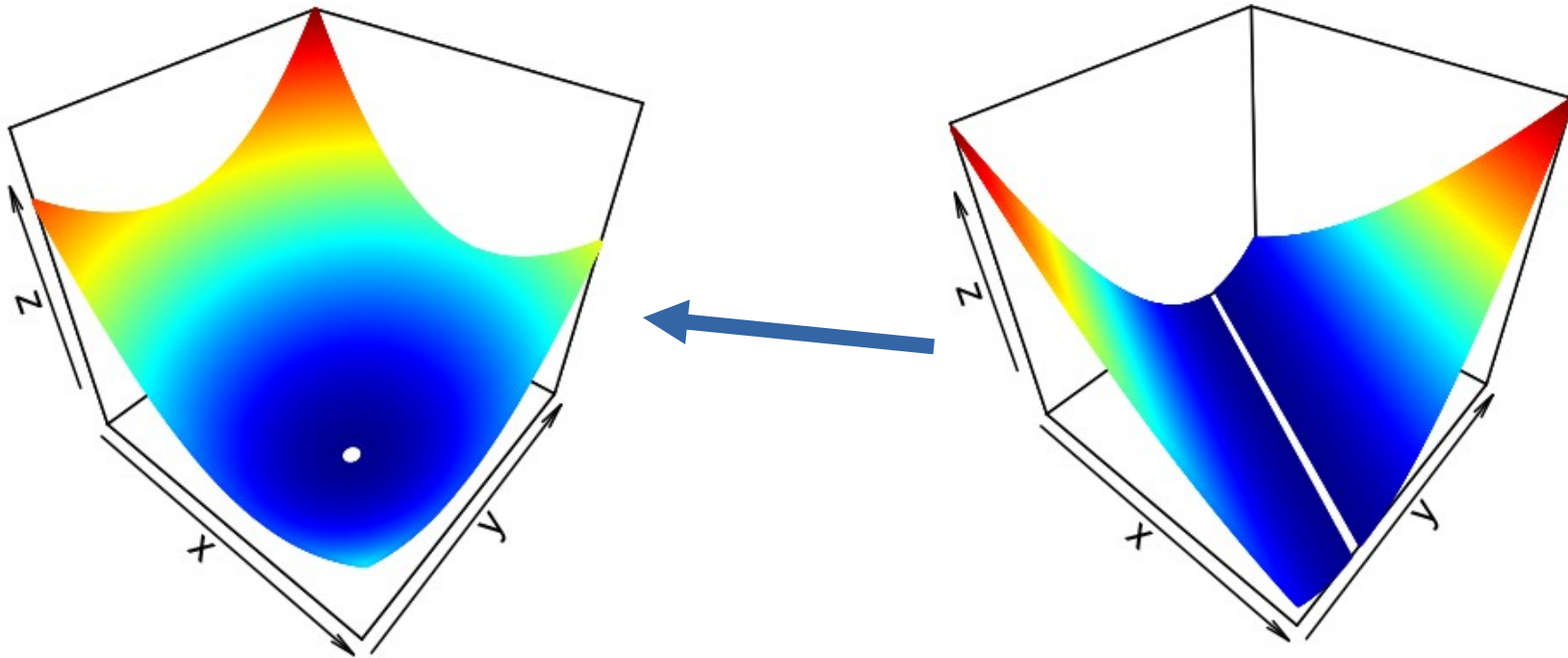
$$\hat{\beta} = (X^T X)^{-1} X^T y$$

$$X^T X$$

The fact that  
the off-diagonal  
> 0, indicates  
that there is  
**collinearity**



# How can we make a model stable?



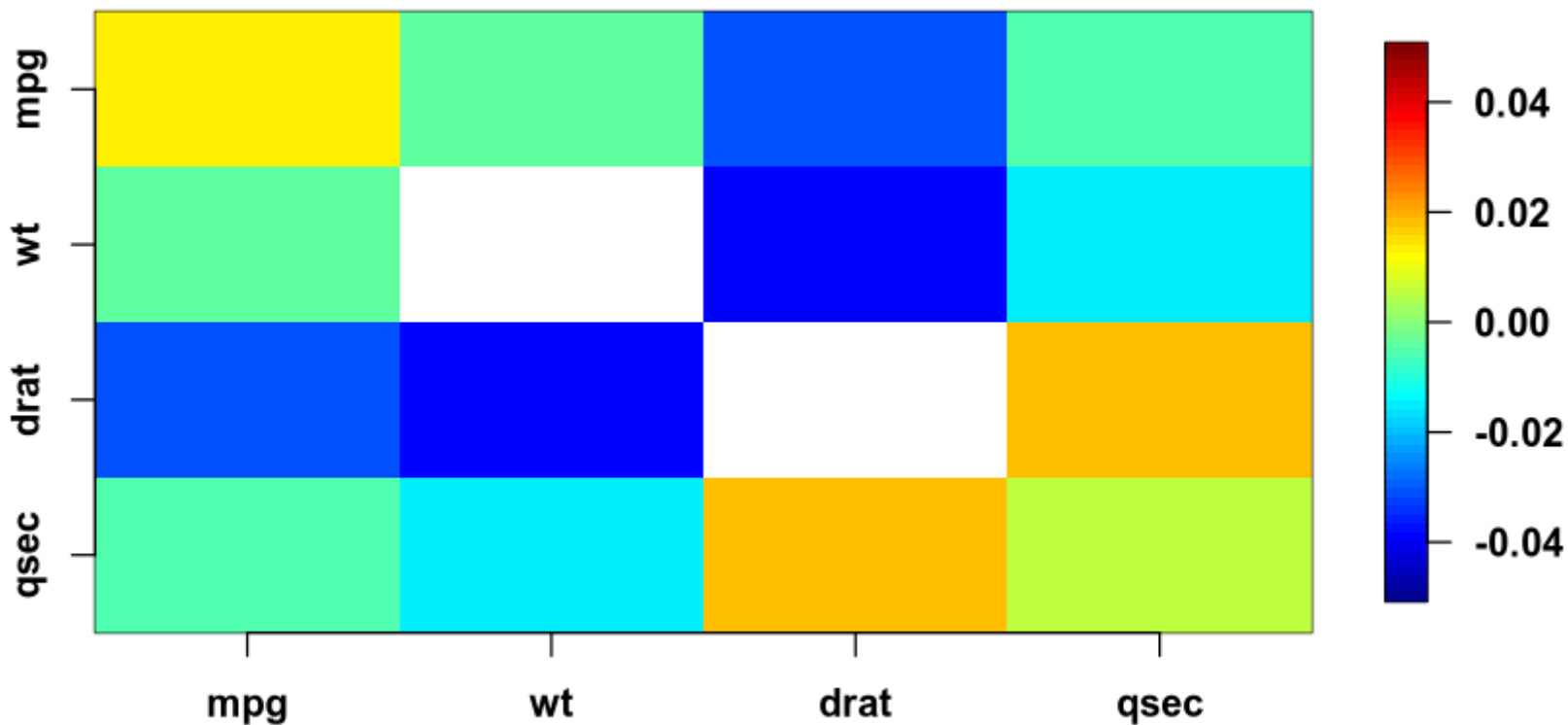
# Regularisation

$$\hat{\beta}_{reg} = (X^T X + \lambda I)^{-1} X^T y$$

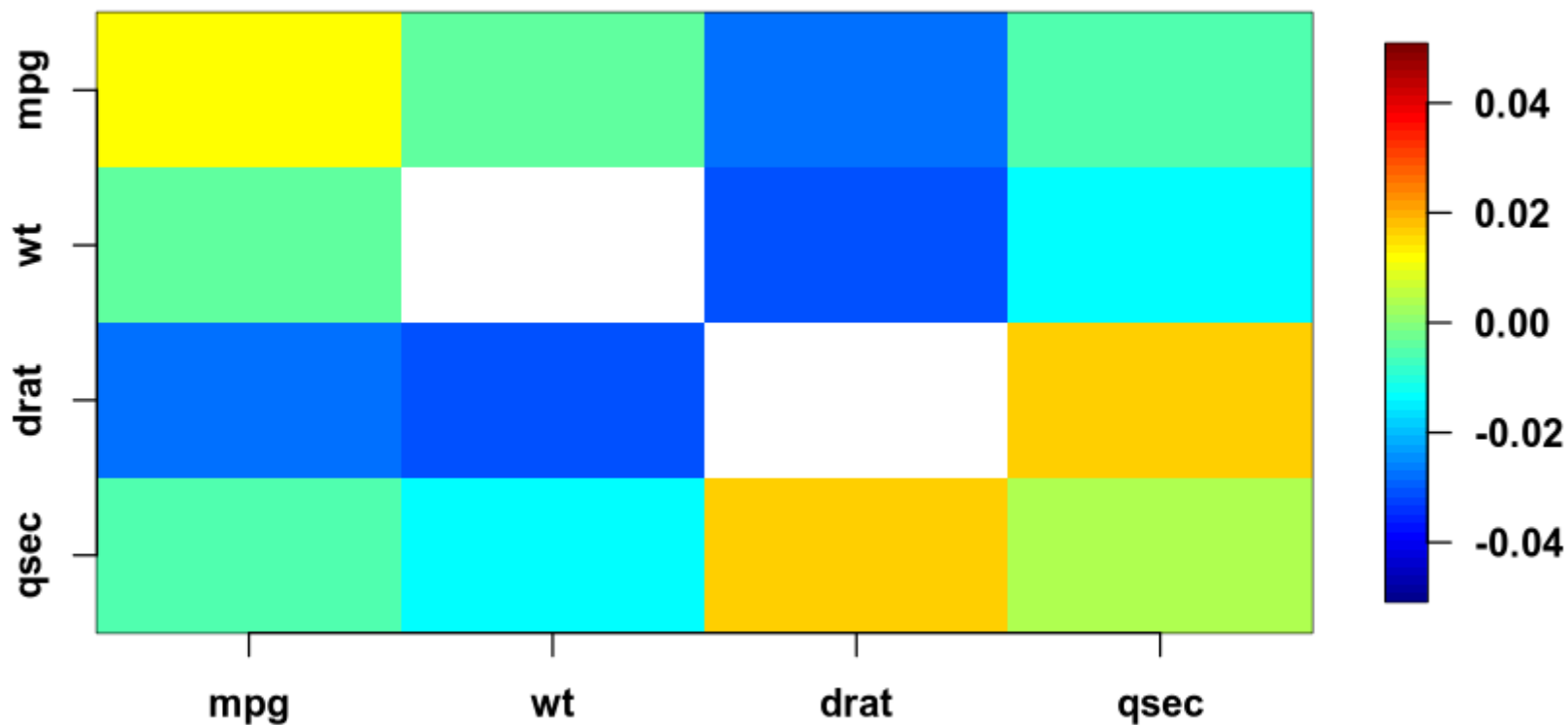
$I$  : an identity matrix with  $p$  predictor variables

$\lambda$  : a constant

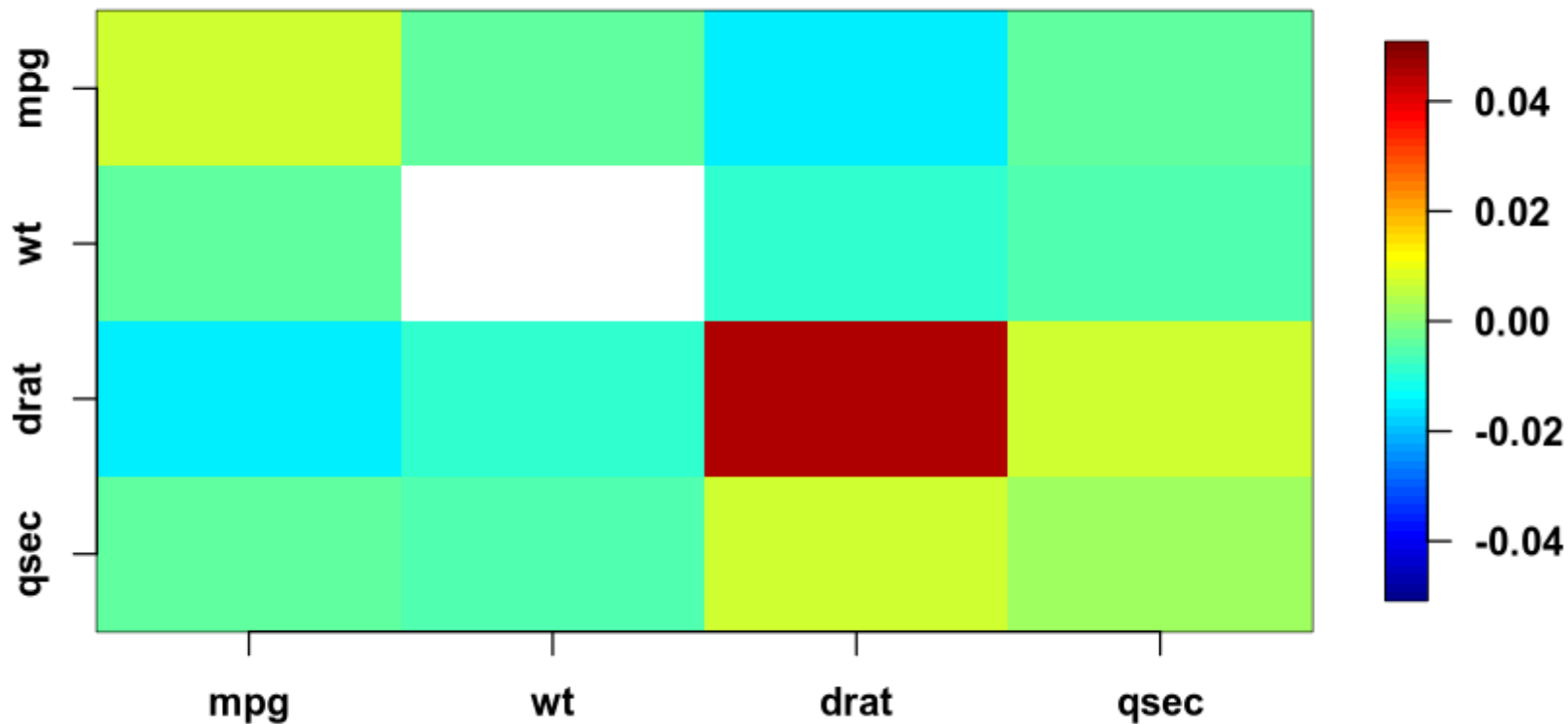
$$(X^T X + \lambda I)^{-1} ; (\lambda = 0)$$



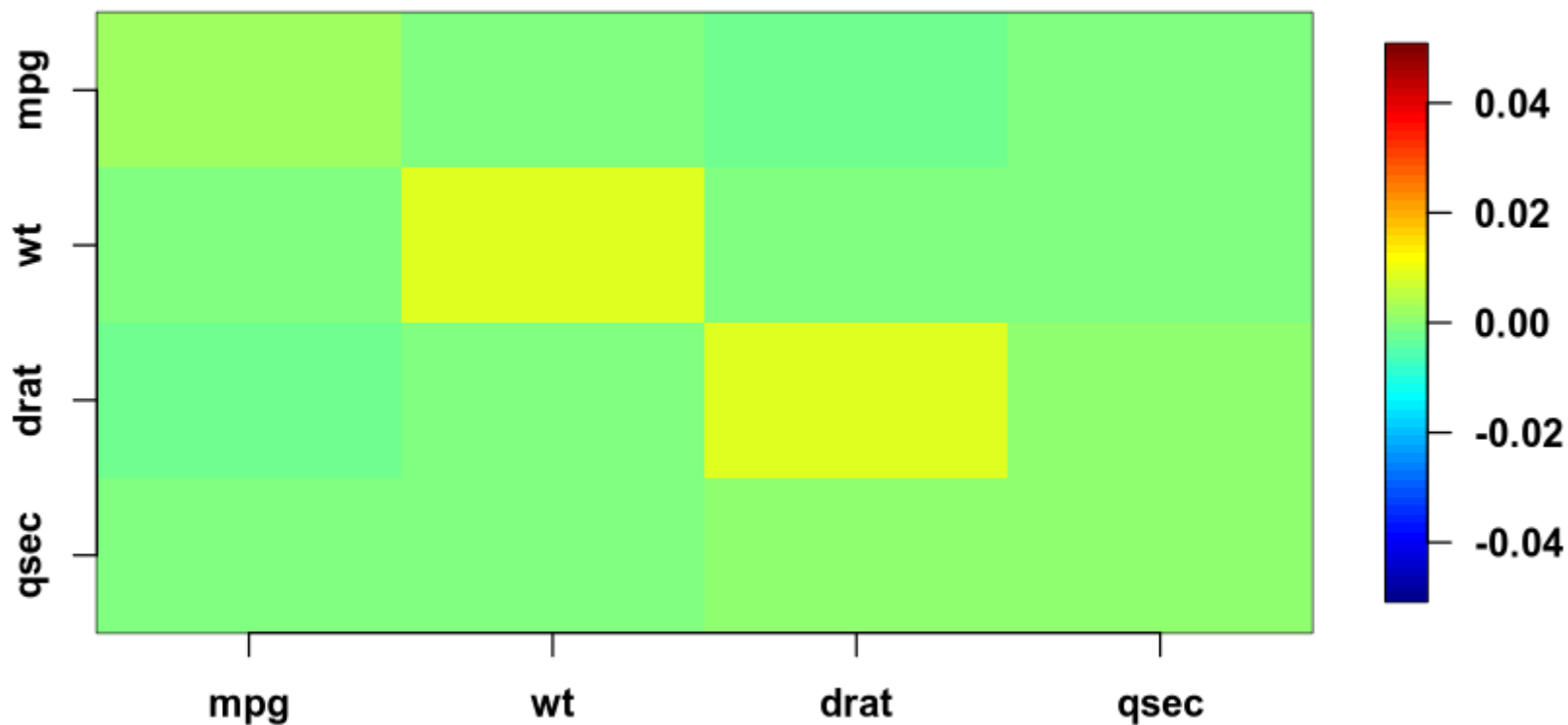
$$(X^T X + \lambda I)^{-1} ; (\lambda = 1)$$



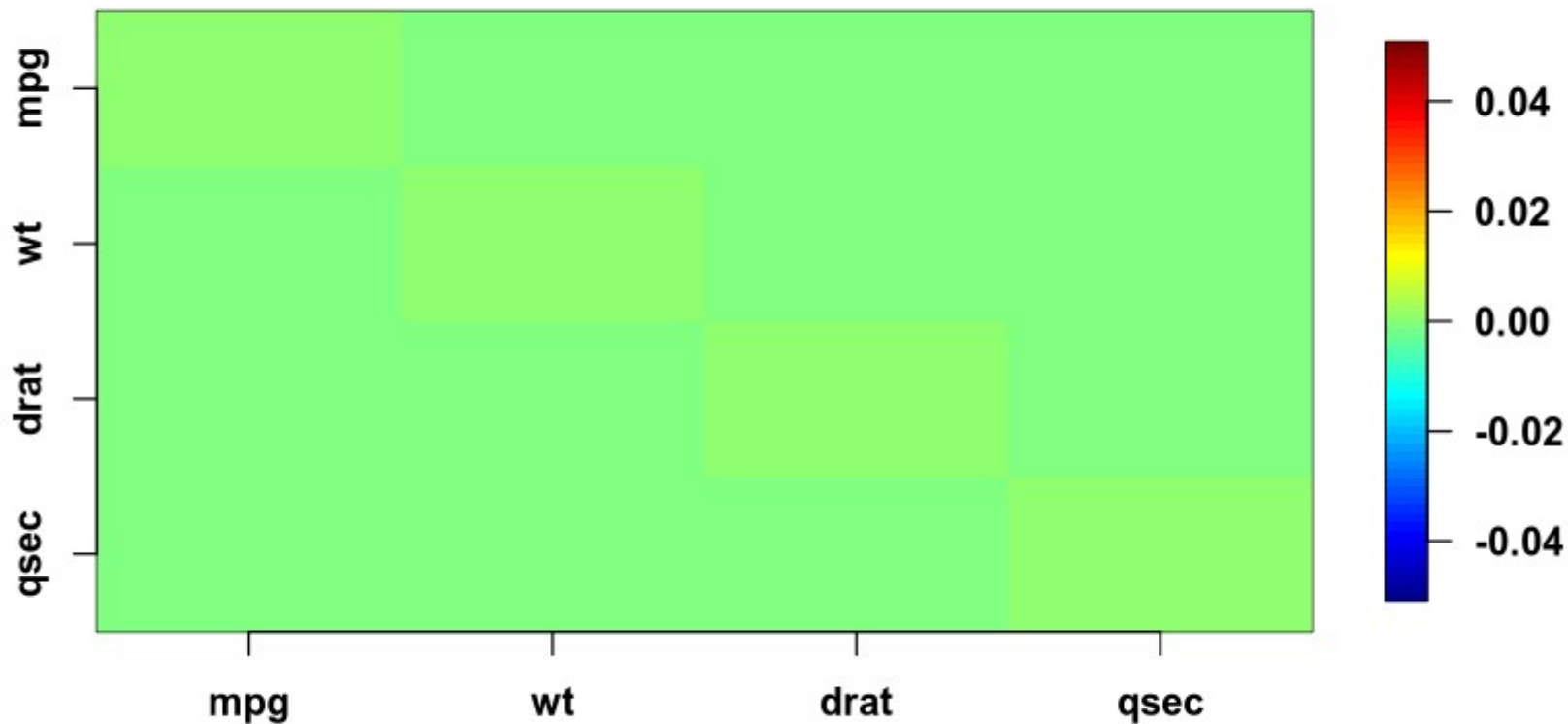
$$(X^T X + \lambda I)^{-1} ; (\lambda = 10)$$



$$(X^T X + \lambda I)^{-1} ; (\lambda = 100)$$

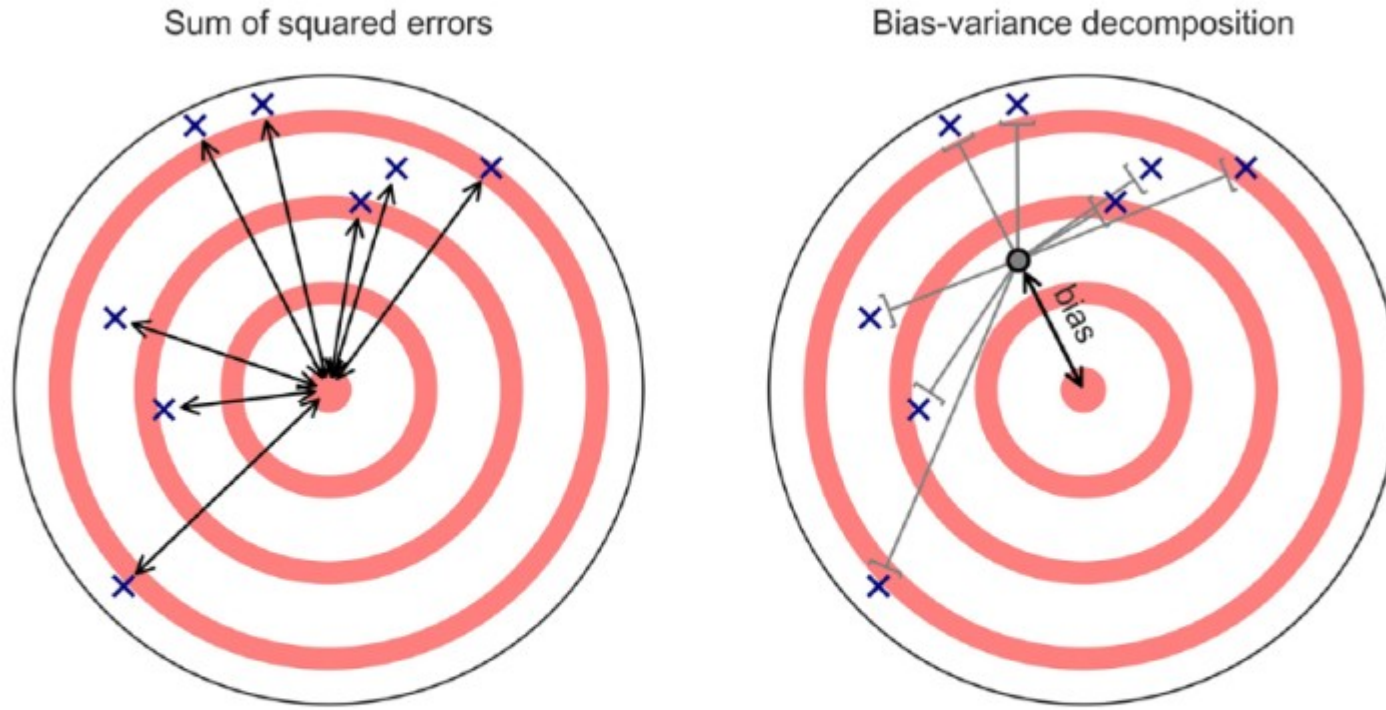


$$(X^T X + \lambda I)^{-1} ; (\lambda = 1000)$$



Regularisation decreases collinearity, but at what price?

I.e. what increases instead?



**Fig. 3.** Schematic illustration of the bias-variance decomposition. (Left) Under the classical error model, prediction error is defined as the sum of squared differences between true scores and observed scores (black lines). (Right) The bias-variance decomposition partitions the total sum of squared errors into two separate components: a bias term that captures a model's systematic tendency to deviate from the true scores in a predictable way (black line) and a variance term that represents the deviations of the individual observations from the model's expected prediction (gray lines).

# Summary

- $R^2$ , AIC and log-likelihood tests can all be adapted to multilevel models
  - However, some debate exists around what are the number of effective parameters
  - And none of them handle collinearity explicitly
- Regularisation gives a handle on collinearity
  - at the cost of introducing bias

# Learning goals and outline

## *Evaluating Generalised linear mixed models*

- 1) Learning tools for comparing models
  - 1) Variance explained
  - 2) Likelihood ratio tests
  - 3) Information criteria
- 2) Bridging to out-of-sample
  - 1) Introducing regularisation

# Next time

- Penalised regression
  - Ridge and lasso regression
- Doing out-of-sample testing
  - Finding optimal lambda
- Bias variance decomposition
  - Increasing one, reduces the other

# The course plan

## *Week 37:*

Lesson 0: What is it all about?

Class 0: Setting up R and Python — and recollection of the general linear model

## *Week 38:*

Lesson 1: Multilevel linear regression

Class 1: Modelling subject level effects – and how do they differ from group level effects?

## *Week 39:*

Lesson 2: Beyond linear regression — link functions

Class 2: What to do when the response variable is not continuous?

## *Week 40:*

Lesson 3: Evaluating generalised linear mixed models — and the spectre of collinearity

Class 3: Code review

## ***Week 41:***

**Lesson 4: Explanation versus prediction**

**Class 4: Summarising mixed models**

## *Week 42:*

no teaching (go fetch potatoes instead)

## *Week 43:*

Lesson 5: Mid-way evaluation and machine learning introduction

Class 5: Getting Python running

## *Week 44:*

Lesson 6: Linear and logistic regression revisited (machine learning)

Class 6: Categorizing responses based on informed guesses

## *Week 45:*

Lesson 7: Model evaluation and hyperparameter tuning

Class 7: Code review

## *Week 46:*

Lesson 8: Dimensionality reduction — principled component analysis (PCA)

Class 8: What to do with very rich data?

## *Week 47:*

Lesson 9: Neural networks

Class 9: Code review

## *Week 48:*

Lesson 10: Unsupervised classification

Class 10: Create video explainers of central concepts (shared resource)

## *Week 49:*

Lesson 0 again: What was it all about? and final evaluation

Class 11: Summarising machine learning

## *Week 50:*

Student presentations:

Class 12: Work on portfolios: Ask anything!

# Reading questions

- Yarkoni and Westfall
  - What is overfitting?
  - What are training and test errors?
  - How are  $p$ -hacking and overfitting related?
  - How may *big data* guard against overfitting?
- Breimann
  - What are the *Data* and *Algorithmic Modeling Cultures*?
  - What is the Rashomon effect?
  - What is the Occam dilemma?